

How Microsoft Edge Saves and Protects Your Passwords

Microsoft Edge uses a strong, Windows-based encryption system to store and retrieve your login credentials safely.



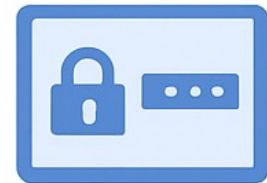
1. Your Windows Login Password: The Root of All Protection

When you sign in to Windows, your password becomes the foundation for protecting encrypted data through the Data Protection API (DPAPI).



2. DPAPI Master Key: Core Encryption Engine

Windows generates a DPAPI master key for each user profile on the system. Microsoft Edge uses this master key to encrypt your passwords before saving them to disk.



3. How Edge Stores Passwords: Encrypted Blobs on Disk

Saved passwords in Microsoft Edge are stored as encrypted blobs – small pieces of protected data that cannot be decrypted without the DPAPI master key. So even if someone copies your browser files or gains access to your system's storage, they still cannot read your saved passwords.



1. Edge retrieves the encrypted blob from disk



2. Edge asks DPAPI to decrypt it



3. The decrypted password appears only in RAM, for just a moment



4. Edge auto-fills the login form